

Industrial Internship Report on "LPG Gas Leakage detector alarm system"

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was my project was LPG Gas Leakage Detector Alarm System using gas sensor technology to detect leakage and provide safety alert

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

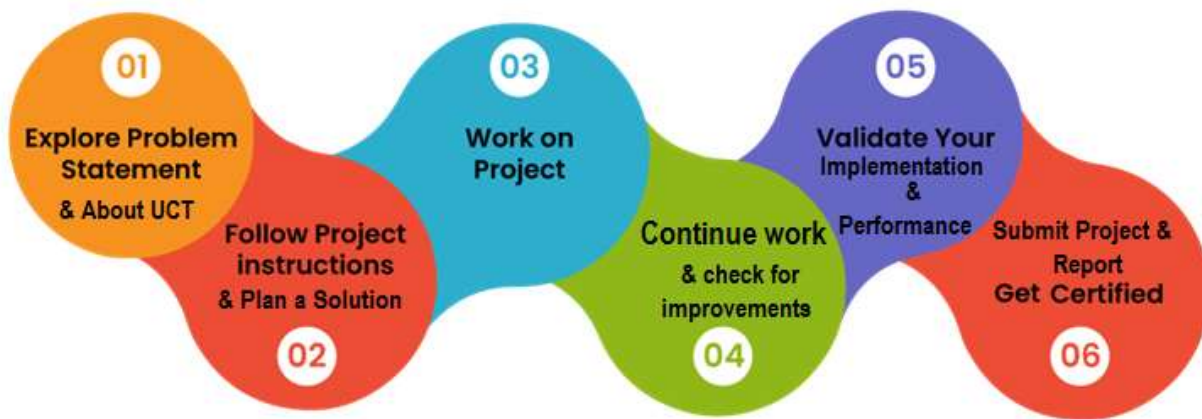
Summary of the whole 6 weeks' work.

About need of relevant Internship in career development.

Brief about Your project/problem statement.

Opportunity given by USC/UCT.

How Program was planned



Your Learnings and overall experience.

Thank to all (with names), who have helped you directly or indirectly.

Your message to your juniors and peers.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoSaWAN), Java Full Stack, Python, Front end** etc.



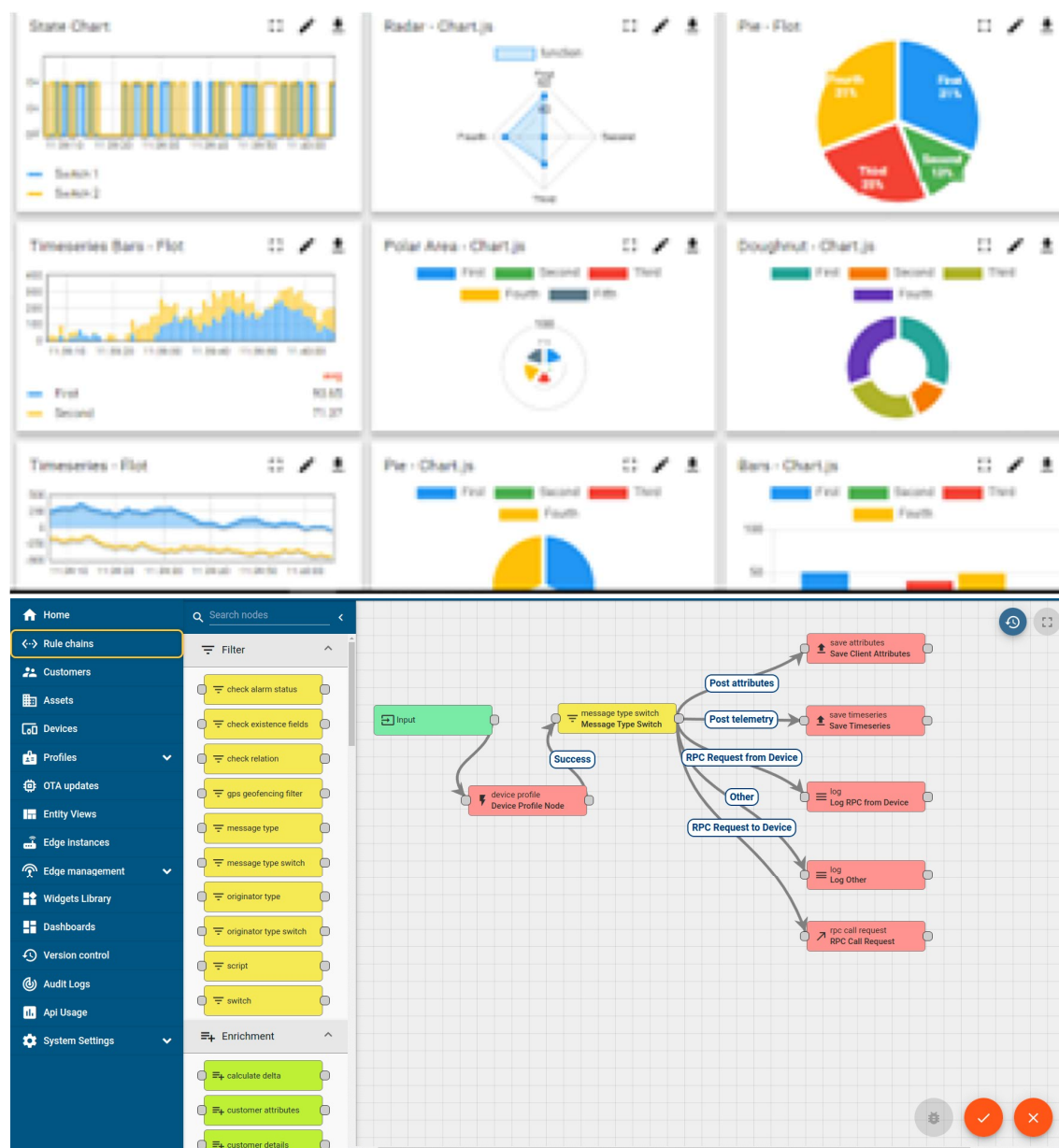
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO040520001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO040520001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



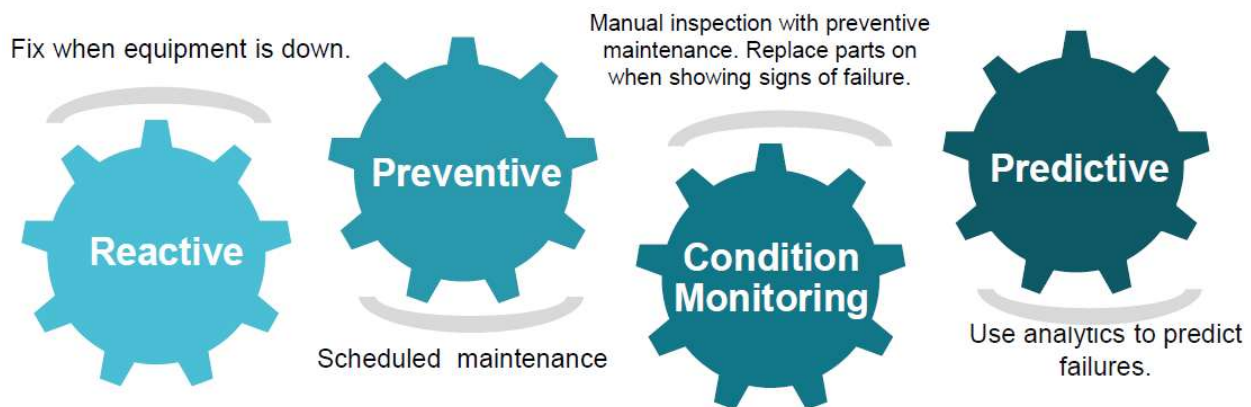


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

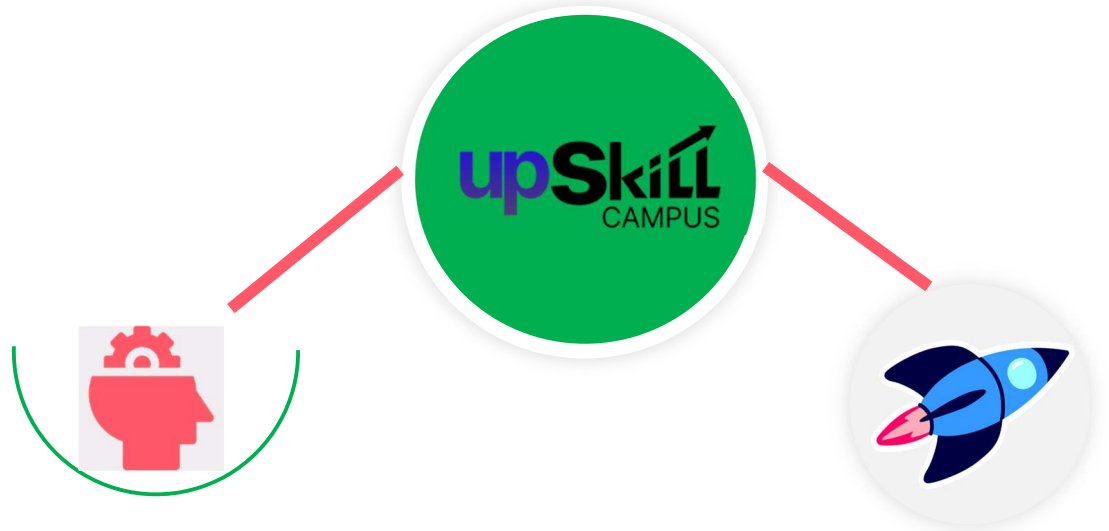
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

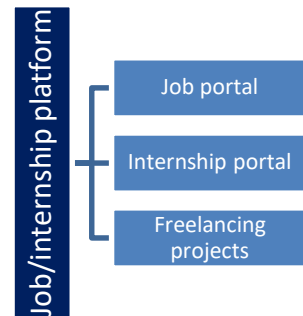
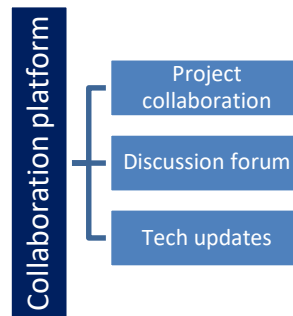
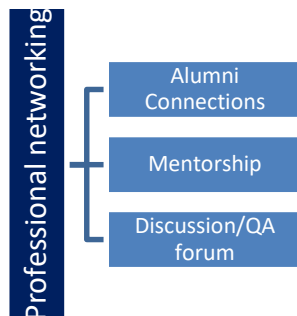
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

[1]

[2]

[3]

2.6 Glossary

Terms	Acronym

3 Problem Statement

In the assigned problem statement

LPG gas leakage can cause dangerous accidents and fire hazards.

This project detects gas leakage using a gas sensor and gives an alarm for safety

4 Existing and Proposed solution

Provide summary of existing solutions provided by others, what are their limitations?

Existing gas leakage systems are expensive and not easily affordable for small users.

What is your proposed solution?

The proposed system uses a gas sensor to detect LPG leakage and provides alarm indication for safety

What value addition are you planning?

This system helps improve safety and can reduce accidents caused by gas leakage

4.1 Code submission (Github link)

<https://github.com/shrutipadyal0212-lgtm/upskillcampus/blob/main/LPGGasLeakageDetector.java>

4.2 Report submission (Github link) : https://github.com/shrutipadyal0212-lgtm/upskillcampus/blob/main/LPGGasLeakageDetector_Shruti_USC_UCT.pdf

5 Proposed Design/ Model

The system uses a gas sensor to detect LPG leakage.

When gas level increases above the limit, the system activates an alarm and warning message for safety.

5.1 High Level Diagram

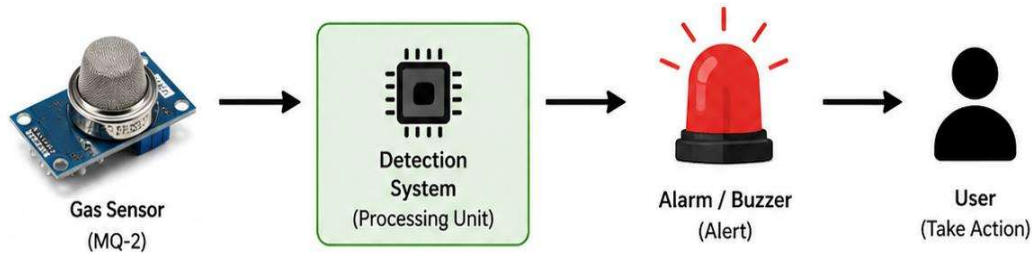
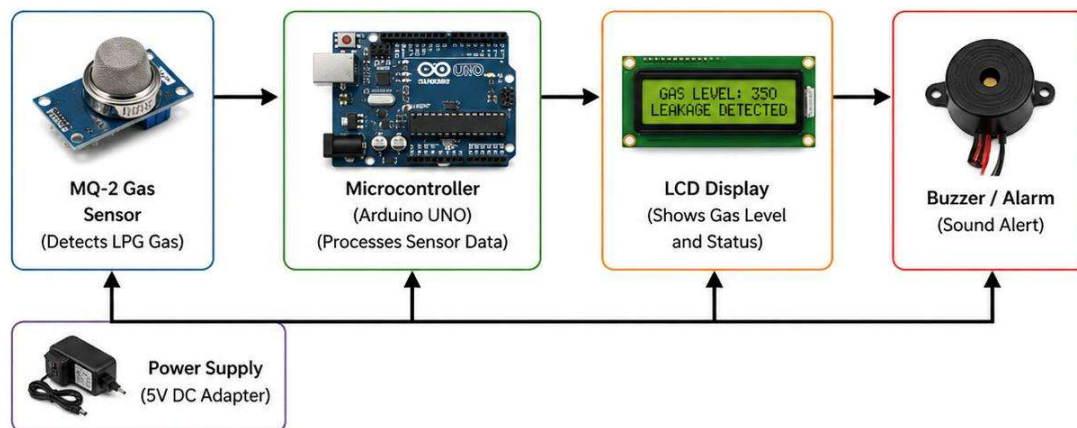


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram



5.3 Interfaces (if applicable)

The LPG Gas Leakage Detection System uses different interfaces to communicate between hardware components. The MQ-2 gas sensor detects LPG gas leakage and sends signals to the Arduino UNO microcontroller. The microcontroller processes the sensor data and displays the gas level on the LCD display. When the gas level exceeds the safety limit, the buzzer alarm is activated to alert the user. The system uses a 5V power supply for proper operation.

6 Performance Test

This is very important part and defines why this work is meant of Real industries, instead of being just academic project.

Here we need to first find the constraints.

How those constraints were taken care in your design?

What were test results around those constraints?

Constraints can be e.g. memory, MIPS (speed, operations per second), accuracy, durability, power consumption etc.

In case you could not test them, but still you should mention how identified constraints can impact your design, and what are recommendations to handle them.

6.1 Test Plan/ Test Cases

Test the MQ-2 sensor with LPG gas.

Check whether the LCD displays gas level correctly.

Verify buzzer activation during gas leakage.

Test system response time.

Check power supply stability

6.2 Test Procedure

Connect all components properly.

Power ON the system.

Bring LPG gas near the MQ-2 sensor.

Observe gas level on LCD display.

Check whether buzzer alarm turns ON.

Remove gas source and observe system reset

6.3 Performance Outcome

The system successfully detected LPG gas leakage and activated the buzzer alarm immediately. The LCD display showed gas status correctly. The project worked efficiently with low power consumption and quick response time

7 My learnings

During this internship project, I learned about LPG gas leakage detection systems and the working of MQ-2 gas sensors. I understood how Arduino UNO interfaces with sensors and output devices like LCD displays and buzzers. I also learned basic programming in Java and gained knowledge about GitHub repository creation, file uploading, and project documentation. This internship improved my technical skills, problem-solving ability, and practical knowledge

8 Future work scope

- Add SMS alert system using GSM module.
- Add mobile app monitoring feature.
- Improve sensor accuracy using advanced sensors.
- Add automatic gas regulator cut-off system.
- Develop wireless IoT-based monitoring system.
- Reduce power consumption for better efficiency